

nating crucial information such as navigation messages, vehicle speed, and warnings.

Are there any broad categories of LCD displays that are witnessing demand?

Yes, reflective LCDs, which use ambient light to reflect in order to read. In 2W cluster applications, where TFT is exposed to direct sunlight, readability is a major issue. Sharp Reflective LCD is a solution as visibility is crystal clear without any

glare and is available in colour too. Equipped with a backlight, it can be used at night also.

Normal TFT has to pump more power through the backlight, which results in more power consumption and backlight life also gets affected to a large extent. This reflective LCD consumes very little power and could be the best fit-in product for the EV segment.

One of the first fears that come to one's mind when we see a large

tablet-like display in cars is of its breaking. But what is the actual risk of these screens breaking?

The market is now shifting to large-size TFT displays in the automotive segment. These displays are automotive-grade LCDs and are tested for shock, vibration, high and low temperature, etc. For more protection and safety, glass bonding is done over TFT. Glass bonding with a cover glass on the LCD protects it from shock, as the hardened adhesive behind the glass acts as a shock

Selection Parameters For LCD Screens For Automotive Applications

Wanting to select the right LCD screen for an automotive design? Here's a list of parameters you need to consider:

High brightness and anti-glare. More brightness and higher contrast make the display clear even during bright sunlight conditions. Manufacturers have enormously brought down surface reflectivity with anti-reflective coatings, making the display easier to pursue during high-sunlight glare conditions. Progressive Super View Technology of Sharp, for example, gives an extremely crisp image under sunlight without pumping more power from the backlight, which helps in maintaining the life and quality of the product.

Screen size and resolution. Auto displays range from 3.8cm to more than 30cm. There's a trend toward bigger display sizes (up to 61cm), with the reception of solid-state and advanced digital clusters. Thus, HD resolution is moving from wide VGA (800 × 480) to 720p and 1080p.

Form factor. In the automotive sector, the most common aspect ratio for display sizes between 17.8cm



Free form display prototype (Credit: Sharp Devices)

and 31.4cm are 16:9, 5:3, 16:9, and 16:6. From hatchback, sedan to SUV, the height varies a lot and, therefore, aspect ratio plays a vital role for display selection. Recent innovations like Free Form help automotive designers to break free from the rectangle to any size or any form factor of the display. The device may be shaped to fulfil a large range of user needs due to the incorporation of IGZO technology and proprietary circuit design methods.

Conventional displays are rectangular because they require a minimal width for the bezel so as to accommodate the drive circuit, called the gate driver, around the perimeter of the screen's display area. With the morpheme display (using IGZO technology), the gate driver's function is dispersed throughout the pixels on the display area. This enables the bezel to be shrunk considerably, and it gives the liberty to style the LCD to match whatever shape the display area of the screen has to be.